

Quadratic Swap Factorization and Completion Geometry in the Cohn–Markoff Monoid

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Abstract

Let

$$X = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix},$$

and let $\mu : \{0, 1\}^* \rightarrow \mathrm{SL}_2(\mathbb{Z})$ be the monoid morphism defined by $\mu(0) = X$ and $\mu(1) = Y$. This representation is classical in the Cohn–Markoff setting, where central and Christoffel words encode Markoff data. We introduce a completion-coordinate description of its word-state dynamics and isolate an exact local rigidity phenomenon.

For $v = (2, 1)^T$ and $q(s) = \mu(s)^T v$, write $F(s) = v^T q(s)$ and $\Psi(s) = (q_1(s), F(s))^T$. A standard unimodular change of coordinates sends the two word append maps to Aigner’s $a = 0$ Cohn generators; we use this normalization to define, for a palindromic base word p with $P = \mu(p)$, $m = F(p)$ and $x = q_1(p)$, the determinant residual

$$R_p(s) = m q_1(s) - x F(s).$$

The main structural statement is the exact adjacent-swap factorization

$$Q_P(q(s10)) - Q_P(q(s01)) = 12F(s)R_p(s),$$

where $Q_P(u) = u^T P u$. Hence the 10/01 swap defect has a strict Farey sign, and its zero set on the word orbit is a single progenitor:

$$Q_P(q(s10)) = Q_P(q(s01)) \iff s = p.$$

Thus the fundamental opposite quadratic return is rigid even without assuming that s is palindromic.

We also introduce relative completion coordinates $\ell(q) = v^T P q$ and $\delta(q) = \det(v, q)$ and prove

$$m Q_P(q) = \ell(q)^2 + \delta(q)^2, \quad \ell(q) + x \delta(q) = m q_2.$$

The homogeneous kernel translation (x, m) in the associated Diophantine completion lattice is transported by the two terminal maps to the quadratic-return translations $(3m^2, m)$ and $(3m^2, -m)$. Finally, we reconstruct an arbitrary centered fiber from reversal data and identify its trace defect exactly. The completion viewpoint was motivated by a line of work beginning with a completion-aware 2:1 deterministic pushdown normalizer and its $p:q$ residual generalization, but the present results are proved here from scratch and do not depend on those manuscripts. They are also independent of the Frobenius uniqueness conjecture; they provide a completion-geometric local structure inside the classical Cohn–Markoff framework.

Keywords. Markoff numbers; Cohn matrices; Christoffel words; Farey tree; Diophantine completion; quadratic forms; unimodular dynamics; palindromes.

1 Introduction

Research provenance and logical independence

The completion viewpoint used in this paper did not originate in Markoff theory. Its starting point was a completion-aware 2:1 deterministic pushdown automaton for the binary balance condition $N_1 = 2N_0$. In that setting, finite control and the pushdown store jointly represented a signed imbalance that had to be cancelled by a future suffix. Passing from the special ratio 2:1 to arbitrary coprime $p:q$ isolates the language-level residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w),$$

and turns end-of-input normalization into the shortest nonnegative solution of a two-variable Diophantine completion equation. The resulting picture separates a canonical representative from translations by a primitive homogeneous kernel and organizes the local completion dynamics by unimodular/Farey data.

The present paper is the next step in that development: it asks what remains of this completion architecture when the underlying object is no longer a pushdown normalizer but the classical Cohn–Markoff matrix monoid. The discovery path is therefore

$ \begin{array}{c} 2:1 \text{ DPDA} \longrightarrow p:q \text{ residual} \longrightarrow \text{shortest completion} \\ \Downarrow \\ \text{Farey/kernel geometry} \longrightarrow \text{Cohn–Markoff quadratic returns} \end{array} $

This paragraph records research provenance only. The article is mathematically self-contained: no theorem from an earlier manuscript is assumed, and no unpublished manuscript is cited in the bibliography.

Classical setting and the present contribution

The classical Markoff equation

$$a^2 + b^2 + c^2 = 3abc$$

is connected to several equivalent binary-tree constructions: Markoff triples, Farey fractions, Christoffel words and Cohn matrices. These correspondences are standard; see, for example, Cohn [2, 3], Aigner [1], Reutenauer [7], and the modern Cohn/Farey treatment in Lagisquet–Pelantová–Tavenas–Vuillon [4]. The Frobenius uniqueness conjecture asks whether a Markoff number can occur as the maximal entry of more than one Markoff triple. That conjecture remains open and is not addressed here as a solved problem.

Two pieces of recent work are especially close to the coordinate language used below. Lagisquet–Pelantová–Tavenas–Vuillon [4] developed simple and composite flip inequalities for a matrix word statistic in a related Cohn-generator normalization. More recently, Veselov [6] observed that Aigner’s $a = 0$ Cohn-matrix indices coincide with Springborn’s Markov fractions [5] and used the same $a = 0$ generators that arise from our unimodular conjugacy. Accordingly, the coordinate conjugacy itself is background here rather than a novelty claim.

The purpose of this paper is different. We study the same matrix monoid through a completion coordinate system and ask for an exact base-dependent quadratic defect. The basic object is the word state

$$q(s) = \mu(s)^T (2, 1)^T.$$

After a fixed unimodular change of coordinates, the two append operations become explicit Farey transformations. For a palindromic base p , its characteristic fraction x/m defines a determinant residual $R_p(s)$ measuring the Farey side of the state $\Psi(s)$. The key point is that this linear

residual controls a quadratic quantity exactly: exchanging the terminal chronology 10 and 01 changes the P -quadratic energy by precisely $12F(s)R_p(s)$.

This factorization separates classical background from the contribution of the present paper. The Cohn tree, Farey tree, Christoffel coding, local word flips, and the $a = 0$ Cohn normalization are not new. What we isolate is a fixed-palindromic-base quadratic sandwich observable whose adjacent 10/01 defect factors exactly into a positive size factor and one Farey determinant. This gives the full magnitude and sign of the defect, a singleton zero set on the word orbit, and a kernel-to-return correspondence. The latter identifies the homogeneous kernel of a two-variable Diophantine completion equation with a fundamental translation in quadratic return coordinates.

The results are algebraic and hold for every palindromic base word, not only for central words. When the base is central, they specialize to the usual Christoffel–Markoff setting. This wider scope is useful because it shows that the local factorization is a property of the Cohn matrix dynamics itself rather than a consequence of the global Markoff tree.

Main results

Let $v = (2, 1)^T$, let

$$q(s) = \mu(s)^T v = (a, b)^T, \quad F(s) = v^T q(s) = 2a + b, \quad \Psi(s) = \begin{pmatrix} a \\ F(s) \end{pmatrix}.$$

For a palindromic word p , put $P = \mu(p)$,

$$m = F(p), \quad x = q_1(p).$$

We prove the following.

- (i) The word-state dynamics is conjugate to the unimodular Farey dynamics

$$\Psi(s0) = \begin{pmatrix} F(s) \\ 3F(s) - q_1(s) \end{pmatrix}, \quad \Psi(s1) = \begin{pmatrix} q_1(s) + 2F(s) \\ 2q_1(s) + 5F(s) \end{pmatrix}.$$

- (ii) The fixed-base determinant residual

$$R_p(s) = mq_1(s) - xF(s) = -\det(\Psi(p), \Psi(s))$$

controls the adjacent chronology swap exactly:

$$Q_P(q(s10)) - Q_P(q(s01)) = 12F(s)R_p(s).$$

Consequently

$$Q_P(q(s10)) = Q_P(q(s01)) \iff s = p.$$

- (iii) In relative coordinates

$$\ell(q) = v^T Pq, \quad \delta(q) = \det(v, q),$$

one has

$$mQ_P(q) = \ell(q)^2 + \delta(q)^2, \quad \ell(q) + x\delta(q) = mq_2.$$

Thus every word state lies on a fixed Diophantine completion lattice.

- (iv) If $(a, f) = \Psi(s)$ and $R = ma - xf$, then

$$(\ell, \delta)(q(s10)) = (3mf + R, f),$$

$$(\ell, \delta)(q(s01)) = (3mf - R, -f).$$

The kernel lift $(a, f) \mapsto (a + x, f + m)$ leaves R fixed and induces

$$(\ell, \delta) \mapsto (\ell, \delta) + (3m^2, m)$$

on the 10 branch and

$$(\ell, \delta) \mapsto (\ell, \delta) + (3m^2, -m)$$

on the 01 branch.

- (v) On a centered fiber $F(s) = m$, the matrix $\mu(s)$ is reconstructed from m , $q_1(s)$ and $q_1(s^R)$, and the defect of the wrapped Markoff trace identity is exactly the reversal defect.

The proofs use only explicit 2×2 algebra, unimodularity and the injectivity of the word-state map established by the terminal decoder below. No form of Markoff uniqueness is assumed.

2 The Cohn word-state system

Define

$$X = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}.$$

Both matrices are symmetric, positive and unimodular. Let

$$\mu : \{0, 1\}^* \longrightarrow \mathrm{SL}_2(\mathbb{Z})$$

be the morphism with $\mu(0) = X$, $\mu(1) = Y$. We write words on the right, so

$$\mu(s0) = \mu(s)X, \quad \mu(s1) = \mu(s)Y.$$

Let

$$v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad q(s) = \mu(s)^T v.$$

Because X and Y are symmetric,

$$q(s0) = Xq(s), \quad q(s1) = Yq(s). \tag{1}$$

Every $q(s)$ is primitive because $X, Y \in \mathrm{SL}_2(\mathbb{Z})$ and v is primitive.

Lemma 2.1 (Exact terminal decoder). *Every word state has slope in the invariant interval*

$$I = \left(1, \frac{5}{2}\right).$$

For a nonempty word, the last letter is determined by the slope:

$$q(s0) : \quad \frac{3}{2} < \frac{q_1}{q_2} < \frac{12}{7},$$

$$q(s1) : \quad \frac{7}{3} < \frac{q_1}{q_2} < \frac{29}{12}.$$

These intervals are disjoint. The corresponding inverse maps are

$$T_0(a, b) = (a - b, 2b - a), \quad T_1(a, b) = (a - 2b, 5b - 2a).$$

Consequently the map $s \mapsto q(s)$ is injective.

Proof. The initial vector $v = (2, 1)^T$ has slope $2 \in I$. If a positive vector has slope $r \in I$, then multiplication by X and Y gives slopes

$$\phi_0(r) = \frac{2r+1}{r+1}, \quad \phi_1(r) = \frac{5r+2}{2r+1}.$$

Both maps are increasing on I , and their endpoint images are

$$\phi_0(I) = \left(\frac{3}{2}, \frac{12}{7}\right), \quad \phi_1(I) = \left(\frac{7}{3}, \frac{29}{12}\right).$$

Thus I is invariant and the last letter is read from the unique terminal interval. Since

$$X^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad Y^{-1} = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix},$$

the displayed inverse maps recover the predecessor state. Iteration reaches the unique empty state v , proving injectivity. $\square$

We use the scalar

$$F(s) = v^T q(s). \tag{2}$$

If $q(s) = (a, b)^T$, then

$$F(s) = 2a + b.$$

Lemma 2.2 (Reversal transpose). *For every word s ,*

$$\mu(s^R) = \mu(s)^T.$$

In particular, if $p = p^R$ is a palindrome, then $P = \mu(p)$ is symmetric.

Proof. Transpose reverses the order of a product, and both generators are symmetric. $\square$

For a palindromic base p , write

$$P = \mu(p) = \begin{pmatrix} A & B \\ B & D \end{pmatrix}, \quad m = F(p) = v^T P v, \tag{3}$$

and

$$x = q_1(p) = (Pv)_1 = 2A + B. \tag{4}$$

Then

$$Pv = \begin{pmatrix} x \\ m - 2x \end{pmatrix}.$$

Lemma 2.3 (Characteristic divisibility). *With the notation above,*

$$x^2 + 1 = Am.$$

In particular

$$x^2 \equiv -1 \pmod{m}.$$

Proof. Since $\det P = AD - B^2 = 1$,

$$\begin{aligned} x^2 + 1 &= (2A + B)^2 + 1 \\ &= 4A^2 + 4AB + B^2 + AD - B^2 \\ &= A(4A + 4B + D) \\ &= Am. \end{aligned}$$

$\square$

The congruence in theorem 2.3 is classical in the Markoff setting. We will use it only as an arithmetic identity attached to a palindromic base.

3 Farey completion coordinates

Define a new coordinate vector by

$$\Psi(s) = \begin{pmatrix} q_1(s) \\ F(s) \end{pmatrix}. \quad (5)$$

If $q(s) = (a, b)^T$, then

$$\Psi(s) = \begin{pmatrix} a \\ 2a + b \end{pmatrix} = Cq(s), \quad C = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

The matrix C is unimodular, so $\Psi(s)$ is primitive whenever $q(s)$ is primitive.

Theorem 3.1 (Word-to-Farey coordinate conjugacy). *For every word s , writing $\Psi(s) = (a, f)^T$, one has*

$$\Psi(s0) = \begin{pmatrix} f \\ 3f - a \end{pmatrix}, \quad (6)$$

and

$$\Psi(s1) = \begin{pmatrix} a + 2f \\ 2a + 5f \end{pmatrix}. \quad (7)$$

Equivalently, for $\rho(s) = a/f$,

$$\rho(s0) = \frac{1}{3 - \rho(s)}, \quad \rho(s1) = \frac{\rho(s) + 2}{2\rho(s) + 5}. \quad (8)$$

Proof. Conjugating the two append maps in equation (1) by C gives

$$CXC^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}, \quad CYC^{-1} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}.$$

Applying these matrices to $(a, f)^T$ gives equations (6) and (7); division of the two coordinates gives equation (8). $\square$

Remark 3.2 (Relation with the $a = 0$ Cohn normalization). *The two conjugated generators*

$$\begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$$

are exactly Aigner's $a = 0$ Cohn generators. Veselov [6] uses this normalization to identify Cohn-matrix indices with Markov fractions. Thus theorem 3.1 is recorded as the coordinate bridge needed for the quadratic results below, not as a new Cohn–Farey correspondence.

Fix from now on a palindromic base p and write

$$\Psi(p) = \begin{pmatrix} x \\ m \end{pmatrix}.$$

Definition 3.3 (Farey residual). *For any word s with $\Psi(s) = (a, f)^T$, define*

$$R_p(s) = ma - xf. \quad (9)$$

Thus

$$R_p(s) = -\det(\Psi(p), \Psi(s)). \quad (10)$$

The sign of $R_p(s)$ records on which side of x/m the fraction a/f lies. In the original q -coordinates,

$$R_p(s) = (m - 2x)q_1(s) - xq_2(s). \quad (11)$$

This residual is the natural linear coordinate for the Diophantine equation

$$mu - xd = r,$$

whose homogeneous kernel is generated by (x, m) . The next section shows that the same completion lattice appears canonically from the quadratic form defined by P .

4 Relative completion coordinates

For $u \in \mathbb{Z}^2$, put

$$Q_P(u) = u^T P u.$$

For a vector $q \in \mathbb{Z}^2$, define

$$\ell(q) = v^T P q, \quad \delta(q) = \det(v, q). \quad (12)$$

Since $v = (2, 1)^T$, if $q = (q_1, q_2)^T$ then

$$\delta(q) = 2q_2 - q_1.$$

Theorem 4.1 (Relative completion coordinates). *For every $q \in \mathbb{Z}^2$,*

$$mQ_P(q) = \ell(q)^2 + \delta(q)^2. \quad (13)$$

Moreover,

$$\ell(q) + x\delta(q) = mq_2, \quad (14)$$

and

$$\ell(q) - (m - x)\delta(q) = m(q_1 - q_2). \quad (15)$$

Consequently

$$\ell(q) \equiv -x\delta(q) \pmod{m}. \quad (16)$$

Proof. The two-dimensional Gram determinant identity gives

$$(v^T P v)(q^T P q) - (v^T P q)^2 = \det(P) \det(v, q)^2.$$

Since $\det P = 1$, this is exactly equation (13).

Using $Pv = (x, m - 2x)^T$ and $\delta(q) = 2q_2 - q_1$,

$$\begin{aligned} \ell(q) + x\delta(q) &= xq_1 + (m - 2x)q_2 + x(2q_2 - q_1) \\ &= mq_2, \end{aligned}$$

which proves equation (14). Similarly,

$$\begin{aligned} \ell(q) - (m - x)\delta(q) &= xq_1 + (m - 2x)q_2 - (m - x)(2q_2 - q_1) \\ &= m(q_1 - q_2). \end{aligned}$$

Reducing equation (14) modulo m gives equation (16). □

Corollary 4.2 (Exact completion lattice). *The image of the integral map $q \mapsto (\ell(q), \delta(q))$ is exactly*

$$\Lambda_{m,x} = \{(\ell, d) \in \mathbb{Z}^2 : \ell + xd \equiv 0 \pmod{m}\}.$$

In particular, this map identifies $\mathbb{Z}^2$ with an index- m sublattice of $\mathbb{Z}^2$.

Proof. In standard coordinates the map has matrix

$$L = \begin{pmatrix} x & m - 2x \\ -1 & 2 \end{pmatrix}, \quad \det L = m.$$

By equation (16), its image is contained in $\Lambda_{m,x}$. The latter has index m , since it is the kernel of the surjective homomorphism $(\ell, d) \mapsto \ell + xd \pmod{m}$. The image of L also has index $|\det L| = m$, so the two lattices coincide. □

Remark 4.3 (Diophantine completion interpretation). *For $d \geq 0$, equation (14) reads*

$$mu - xd = \ell, \quad u = q_2.$$

For each fixed value of ℓ , the corresponding slice of $\Lambda_{m,x}$ is the integer solution set of this two-variable completion equation, whose homogeneous kernel is generated by (x, m) .

The sign of δ also remembers the terminal branch.

Lemma 4.4 (Terminal sign). *For every word s ,*

$$\delta(q(s0)) = q_2(s) > 0, \quad \delta(q(s1)) = -q_1(s) < 0.$$

Proof. Use equation (1) and compute

$$\delta(X(a, b)^T) = b, \quad \delta(Y(a, b)^T) = -a.$$

All word states have positive coordinates. □

Proposition 4.5 (Palindromic sandwich identity). *For every word t and every palindromic base p ,*

$$\boxed{Q_P(q(t)) = F(tpt^R)}. \quad (17)$$

Moreover,

$$F(w) = \mu(0w1)_{12} \quad (18)$$

for every word w . Hence the quadratic observable is also an ordinary Cohn matrix-entry statistic of a palindromic sandwich.

Proof. Since $p = p^R$, one has

$$\mu(tpt^R) = \mu(t)P\mu(t)^T,$$

which is symmetric. Therefore

$$F(tpt^R) = v^T \mu(tpt^R)^T v = v^T \mu(t)P\mu(t)^T v = Q_P(q(t)).$$

For equation (18), write $\mu(w) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$. A direct multiplication gives

$$\mu(0w1)_{12} = 4A + 2B + 2C + D = v^T \mu(w)^T v = F(w).$$

□

5 Quadratic chronology and Farey curvature

We now compare the two opposite two-letter chronologies 10 and 01. If $q(s) = q$, then

$$q(s10) = XYq, \quad q(s01) = YXq.$$

Theorem 5.1 (Exact adjacent-swap factorization). *Let p be a palindromic base, $P = \mu(p)$, and define m, x as above. Then for every word s ,*

$$\boxed{Q_P(q(s10)) - Q_P(q(s01)) = 12F(s)R_p(s)}. \quad (19)$$

Equivalently,

$$Q_P(q(s10)) - Q_P(q(s01)) = -12F(s) \det(\Psi(p), \Psi(s)). \quad (20)$$

Proof. Write

$$P = \begin{pmatrix} A & B \\ B & D \end{pmatrix}, \quad q = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix}.$$

A direct multiplication gives

$$(XY)^T P(XY) - (YX)^T P(YX) = \begin{pmatrix} 48B + 24D & -24A + 6D \\ -24A + 6D & -24A - 12B \end{pmatrix}.$$

The associated quadratic polynomial factors as

$$\begin{aligned} q^T ((XY)^T P(XY) - (YX)^T P(YX)) q \\ = 12(2q_1 + q_2)((2B + D)q_1 - (2A + B)q_2). \end{aligned}$$

Now $2q_1 + q_2 = F(s)$, $x = 2A + B$, and

$$2B + D = m - 2x.$$

Therefore the second factor is

$$(m - 2x)q_1 - xq_2 = mq_1 - xF(s) = R_p(s),$$

which proves equation (19). Equation (20) follows from equation (10). $\square$

Corollary 5.2 (Word-level swap factorization). *For every word s ,*

$$F(s10p01s^R) - F(s01p10s^R) = 12F(s)R_p(s).$$

Equivalently, after the standard $0(\cdot)1$ wrapping, the same identity is a difference of upper-right Cohn matrix entries.

Proof. Apply theorem 4.5 with $t = s10$ and $t = s01$, noting that $(s10)^R = 01s^R$ and $(s01)^R = 10s^R$. The wrapped statement follows from equation (18). $\square$

Remark 5.3 (Relation to flip methods). *Lagisquet–Pelantová–Tavenas–Vuillon [4] obtain sign and equality rules for simple and composite flips of a related matrix-word m -value. The flip idea itself is therefore classical background. The point of theorems 5.1 and 5.2 is the closed base-dependent factorization of the palindromic-sandwich defect into $12F(s)$ times a single determinant residual.*

Corollary 5.4 (Farey sign of the swap). *The sign of the quadratic chronology defect is the sign of*

$$\frac{q_1(s)}{F(s)} - \frac{x}{m}.$$

More precisely,

$$Q_P(q(s10)) > Q_P(q(s01)) \iff \frac{q_1(s)}{F(s)} > \frac{x}{m},$$

and the inequality reverses on the other side.

Proof. Since $F(s), m > 0$,

$$R_p(s) = mF(s) \left(\frac{q_1(s)}{F(s)} - \frac{x}{m} \right).$$

Apply theorem 5.1. $\square$

The zero set is rigid because both Farey coordinate vectors are primitive.

Theorem 5.5 (Zero-curvature return rigidity). *For every word s ,*

$$\boxed{Q_P(q(s10)) = Q_P(q(s01)) \iff s = p.} \quad (21)$$

No palindromicity hypothesis on s is required.

Proof. By theorem 5.1, equality is equivalent to $R_p(s) = 0$, since $F(s) > 0$. Thus $\Psi(s)$ is rationally parallel to $\Psi(p)$. Both are primitive integer vectors with positive coordinates. Hence the rational proportionality factor is positive, and primitivity forces it to be 1. Therefore

$$\Psi(s) = \Psi(p).$$

Because C is unimodular, this gives $q(s) = q(p)$. By the exact decoder in theorem 2.1, equality of word states implies equality of the underlying words. Hence $q(s) = q(p)$ implies $s = p$. The converse is immediate from $R_p(p) = 0$. $\square$

Example 5.6 (The base $p = 0$). *Take $p = 0$. Then*

$$P = X, \quad (x, m) = (5, 13), \quad x^2 + 1 = 2m,$$

and

$$R_p(s) = 13q_1(s) - 5F(s).$$

The first few values illustrate the exact sign and zero laws:

s	$F(s)$	$R_p(s)$	$Q_P(q(s10)) - Q_P(q(s01))$
ε	5	1	60
0	13	0	0
1	29	11	3828
00	34	-1	-408

The unique zero in these rows occurs at the base progenitor $s = p = 0$, as theorem 5.5 predicts.

Remark 5.7 (Local, not global, Markoff rigidity). *Theorem 5.5 is a same-progenitor statement. It does not assert that two arbitrary distinct Christoffel words with the same Markoff number have a common progenitor to which the theorem can be applied. Thus it is independent of, and does not prove, the Frobenius uniqueness conjecture.*

6 Kernel lifts and quadratic returns

The preceding two coordinate systems are related more tightly than the common residual notation suggests. We first express the relative quadratic coordinates after the two terminal chronologies directly in terms of $\Psi(s)$.

Theorem 6.1 (Terminal transport formula). *Let*

$$\Psi(s) = \begin{pmatrix} a \\ f \end{pmatrix}, \quad R = ma - xf.$$

Then

$$(\ell, \delta)(q(s10)) = (3mf + R, f), \quad (22)$$

and

$$(\ell, \delta)(q(s01)) = (3mf - R, -f). \quad (23)$$

Proof. Write $q(s) = (a, b)^T$, so $f = 2a + b$. Directly from XYq and YXq one obtains

$$\delta(XYq) = f, \quad \delta(YXq) = -f.$$

For the first coordinates, substitute $P = \begin{pmatrix} A & B \\ B & D \end{pmatrix}$, $m = 4A + 4B + D$ and $x = 2A + B$ into $\ell = v^T P(\cdot)$; simplification gives

$$\begin{aligned} \ell(XYq) &= 3mf + (ma - xf), \\ \ell(YXq) &= 3mf - (ma - xf). \end{aligned}$$

These are equations (22) and (23). □

Now consider the Diophantine completion equation

$$ma - xf = R. \tag{24}$$

Its integer solution set is invariant under the primitive kernel translation

$$(a, f) \mapsto (a + x, f + m). \tag{25}$$

Theorem 6.2 (Kernel lift equals quadratic-return translation). *Fix a residual R . Under the terminal 10 transport, the kernel lift equation (25) induces*

$$(\ell, \delta) \mapsto (\ell, \delta) + (3m^2, m). \tag{26}$$

Under the terminal 01 transport it induces

$$(\ell, \delta) \mapsto (\ell, \delta) + (3m^2, -m). \tag{27}$$

Proof. The kernel lift leaves R invariant:

$$m(a + x) - x(f + m) = ma - xf = R.$$

Apply theorem 6.1. On the 10 branch,

$$(3m(f + m) + R, f + m) - (3mf + R, f) = (3m^2, m).$$

On the 01 branch,

$$(3m(f + m) - R, -f - m) - (3mf - R, -f) = (3m^2, -m).$$

□

Remark 6.3 (Ambient lattice versus word orbit). *The kernel translate $(a, f) \mapsto (a + x, f + m)$ is an operation on the ambient Diophantine completion lattice. A translated lattice point need not itself belong to the Cohn word orbit. The theorem is therefore an exact lattice-transport identity; theorem 6.4 records the distinguished return points that do arise from the base word.*

The base state itself has $\Psi(p) = (x, m)$ and $R_p(p) = 0$; in particular, $\Psi(p)$ is the primitive homogeneous kernel vector. Applying the two terminal transports to this distinguished kernel point gives the fundamental blocks $p10$ and $p01$.

Corollary 6.4 (Fundamental quadratic returns). *For the two blocks*

$$C_1 = p10, \quad C_0 = p01,$$

one has

$$(\ell, \delta)(q(C_1)) = (3m^2, m), \quad (\ell, \delta)(q(C_0)) = (3m^2, -m). \tag{28}$$

Consequently

$$m \mid \ell(q(C_i)), \quad m \mid \delta(q(C_i)), \quad i = 0, 1.$$

Proof. Set $(a, f) = (x, m)$ and $R = 0$ in theorem 6.1. □

This is the promised kernel/return correspondence: a homogeneous translation in the completion lattice becomes a fixed Euclidean translation in the relative quadratic plane.

7 Characteristic roots and Farey parents

The characteristic congruence in theorem 2.3 determines the oriented Farey parents of the pair (m, x) explicitly.

Theorem 7.1 (Characteristic root determines the oriented Farey parents). *Assume*

$$m > x > 0, \quad \gcd(m, x) = 1, \quad x^2 + 1 = km.$$

Then

$$\boxed{(c, u) = (x, k), \quad (v, d) = (m - x, x - k)} \quad (29)$$

satisfy

$$(m, x) = (c, u) + (v, d),$$

and the oriented unit-determinant relations

$$mu - xc = 1, \quad xv - md = 1.$$

Moreover, they are the unique such decomposition with $0 < c < m$ and $0 < v < m$. (The boundary case $(m, x) = (2, 1)$ has $d = 0$; otherwise the displayed parent coordinates are positive.) Hence the two unit-residual lifts are

$$(k, x), \quad (-(x - k), -(m - x)). \quad (30)$$

Proof. The characteristic identity gives $x^2 - km = -1$. Therefore the displayed first pair satisfies

$$mk - x \cdot x = 1.$$

For the second pair,

$$x(m - x) - m(x - k) = mk - x^2 = 1,$$

and their sum is (m, x) . Thus the displayed pairs have the required oriented unit-determinant relations.

For uniqueness, any normalized first parent satisfies $xc \equiv -1 \pmod{m}$. Since $x^2 \equiv -1 \pmod{m}$ and x is invertible modulo m , one gets $c \equiv x \pmod{m}$. The condition $0 < c < m$ forces $c = x$, and then $u = k$. Similarly $xv \equiv 1 \pmod{m}$ gives $v \equiv -x \equiv m - x \pmod{m}$; the condition $0 < v < m$ forces $v = m - x$, and then $d = x - k$. Finally, $k = (x^2 + 1)/m \leq x$, with equality only for $(m, x) = (2, 1)$, which proves the stated positivity qualification. $\square$

Remark 7.2. The arithmetic congruence $x^2 \equiv -1 \pmod{m}$ and the classical Farey/Cohn correspondence are part of the established Markoff literature. The role of theorem 7.1 here is to identify the characteristic data with the same completion kernel used in theorem 6.2, so that the local Farey and quadratic-return descriptions share one coordinate system.

8 Centered fibers and reversal defects

The completion viewpoint also gives a compact reconstruction statement on a fixed F -fiber. This part does not require the base word to be palindromic.

Theorem 8.1 (Centered-fiber reconstruction). *Let s be any word with $F(s) = m$, and write*

$$\mu(s) = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

$$r = q_1(s) = 2A + C, \quad r' = q_1(s^R) = 2A + B.$$

Then

$$rr' + 1 = Am. \quad (31)$$

Thus, with

$$K = \frac{rr' + 1}{m},$$

the full matrix is reconstructed from (m, r, r') as

$$\mu(s) = \begin{pmatrix} K & r' - 2K \\ r - 2K & m - 2r - 2r' + 4K \end{pmatrix}. \quad (32)$$

Proof. Since $F(s) = v^T \mu(s)^T v$,

$$m = 4A + 2B + 2C + D.$$

Also $\det \mu(s) = AD - BC = 1$. Hence

$$\begin{aligned} rr' + 1 &= (2A + C)(2A + B) + 1 \\ &= 4A^2 + 2AB + 2AC + BC + AD - BC \\ &= A(4A + 2B + 2C + D) \\ &= Am. \end{aligned}$$

Thus $A = K$. The identities $r = 2A + C$ and $r' = 2A + B$ give $C = r - 2K$ and $B = r' - 2K$, and the formula for D follows from the equation for m . $\square$

Theorem 8.2 (Wrapped trace defect equals reversal defect). *Under the hypotheses of theorem 8.1, let*

$$H = \mu(0s1) = X\mu(s)Y.$$

Then

$$H_{12} = F(s) = m, \quad (33)$$

and

$$\boxed{\text{Tr}(H) - 3H_{12} = r - r'}. \quad (34)$$

Consequently

$$\boxed{\text{Tr}(\mu(0s1)) = 3\mu(0s1)_{12} \iff s = s^R}. \quad (35)$$

Proof. A direct multiplication of $X\mu(s)Y$ gives $H_{12} = 4A + 2B + 2C + D = m$ and

$$\text{Tr}(H) - 3H_{12} = C - B.$$

But

$$C - B = (r - 2A) - (r' - 2A) = r - r'.$$

Thus the trace identity is equivalent to $r = r'$. By theorem 8.1, $r = r'$ is equivalent to $B = C$, hence to symmetry of $\mu(s)$. By theorem 2.2,

$$\mu(s)^T = \mu(s^R).$$

Hence $\mu(s) = \mu(s^R)$, so in particular $q(s) = q(s^R)$. Injectivity of the word-state map from theorem 2.1 gives $s = s^R$. $\square$

9 Relation to the classical Markoff setting

When p is a central palindrome, the word $0p1$ is the corresponding lower Christoffel word, and the matrices above lie in the standard Cohn–Markoff parametrization. The classical theory identifies Christoffel/Farey data with Markoff triples and studies the resulting order and uniqueness questions [1, 7, 4]. The results of this paper add a local completion layer to that picture.

There are three distinct structures:

- (a) the classical tree coordinate, which locates a Christoffel/Cohn object by a Farey fraction;
- (b) the completion residual $R_p(s)$, which measures the determinant displacement of an arbitrary word state from a fixed palindromic base;
- (c) the quadratic chronology defect, which compares the two opposite terminal orders 10 and 01 .

Theorem 5.1 identifies (b) and (c) exactly. In particular, the swap energy has no additional nonlinear zero set: on the word orbit, zero curvature is exactly the base state itself. Theorem 6.2 then identifies the completion kernel with the quadratic return lattice.

This local rigidity should not be confused with global Markoff injectivity. In a global collision, the two competing branches need not have the same progenitor, and the present paper does not supply the missing global core-identification step. The significance here is instead that once a common progenitor is fixed, the opposite chronology is completely controlled by a single Farey determinant.

10 Conclusion

Using the classical $a = 0$ Cohn coordinate normalization, we have isolated a base-dependent quadratic completion geometry on the Cohn word orbit. The main factorization

$$Q_P(q(s10)) - Q_P(q(s01)) = 12F(s)R_p(s)$$

turns a nonlinear-looking comparison of two matrix words into a linear Farey determinant. Its zero set yields the return-rigidity theorem

$$Q_P(q(s10)) = Q_P(q(s01)) \iff s = p.$$

In parallel, the relative coordinates (ℓ, δ) convert the P -quadratic form into an ordinary sum of two squares and place all word states on a fixed completion lattice. The primitive homogeneous kernel of that lattice is carried by the two terminal maps to the fundamental quadratic translations

$$(3m^2, m), \quad (3m^2, -m).$$

The centered-fiber reconstruction and trace-defect identity provide a second, independent expression of the same theme: reversal symmetry is measured by an explicit integral defect rather than by an implicit word condition.

These statements are self-contained consequences of the Cohn matrix monoid and do not depend on a resolution of the Frobenius uniqueness conjecture. They show that Diophantine completion arithmetic and quadratic-return geometry are two coordinate views of the same local unimodular dynamics.

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References

- [1] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013. doi:10.1007/978-3-319-00888-2.
- [2] H. Cohn, "Approach to Markoff's minimal forms through modular functions," *Annals of Mathematics* 61 (1955), 1–12. doi:10.2307/1969618.
- [3] H. Cohn, "Markoff forms and primitive words," *Mathematische Annalen* 196 (1972), 8–22.
- [4] C. Lagisquet, E. Pelantová, S. Tavenas and L. Vuillon, "On the Markov numbers: fixed numerator, denominator, and sum conjectures," *Advances in Applied Mathematics* 130 (2021), 102227. doi:10.1016/j.aam.2021.102227.
- [5] B. Springborn, "The worst approximable rational numbers," *Journal of Number Theory* 263 (2024), 153–205. doi:10.1016/j.jnt.2024.04.013.
- [6] A. P. Veselov, "Markov fractions and Cohn matrices," arXiv:2604.17401, 2026. doi:10.48550/arXiv.2604.17401.
- [7] C. Reutenauer, *From Christoffel Words to Markoff Numbers*, 2nd ed., Oxford University Press, Oxford, 2026. doi:10.1093/9780197907535.001.0001.